ML_Assignment_3

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```
library(e1071)
bank <- read.csv("~/Downloads/UniversalBank.csv")

names(bank)
```

```
##  [1] "ID"                "Age"               "Experience"
##  [4] "Income"            "ZIP.Code"          "Family"
##  [7] "CCAvg"             "Education"         "Mortgage"
## [10] "Personal.Loan"     "Securities.Account" "CD.Account"
## [13] "Online"            "CreditCard"
```

```
# The Three Columns Needed
bank <- bank[, c("Personal.Loan", "Online", "CreditCard")]
# Shorten Column Names
names(bank) <- c("Loan", "Online", "CC")
# Count Missing Values
colSums(is.na(bank))
```

```
##   Loan Online     CC
##      0      0      0
```

```
# Treat Loan as a Category (0=No, 1=Yes)
bank$Loan <- factor(bank$Loan, levels = c(0, 1))
# Treat Online as a Category (0=No, 1=Yes)
bank$Online <- factor(bank$Online, levels = c(0, 1))
# Treat CC as a Category (0=No, 1=Yes)
bank$CC <- factor(bank$CC, levels = c(0, 1))
# Display # of Customers
nrow(bank)
```

```
## [1] 5000
```

```
# Count Customers who Declined vs Accepted the Loan
table(bank$Loan)
```

```
##
##    0    1
## 4520  480
```

0 missing values for Loan, Online, or CC

5000 total rows in dataset

4520 customers declined the loan, while 480(9.6%) accepted this loan

```
# Random Split that is repeatable
set.seed(123)

# Randomly select 60% of customer rows for training
# Replace = FALSE only lets all customers appear 1x
training_rows <- sample(1:nrow(bank), size = 0.60 * nrow(bank), replace = FALSE)

# Create the training dataset from selected rows
train <- bank[training_rows, ]

# Create the validation dataset from all other rows
validation <- bank[-training_rows, ]

# Display the size of each dataset
c(Training = nrow(train), Validation = nrow(validation))
```

```
##   Training Validation
##       3000       2000
```

3000 customers for training and 2000 customers for validation

```
## Part A. Three Way Pivot Table :

# Count customers for every combination of CC, Loan, and Online.
counts <- table(CC = train$CC,
                Loan = train$Loan,
                Online = train$Online)

# Display CC and Loan as rows, with Online as columns.
ftable(counts,
       row.vars = c("CC", "Loan"),
       col.vars = "Online")
```

```
##         Online    0    1
## CC Loan
## 0  0            785 1145
##    1             65  122
## 1  0            317  475
##    1             34   57
```

1 means that the customer either has a bank credit card, accepted the loan, or used online banking

```
## Part B. Conditional Probability :

# Count customers with CC = 1 and Online = 1 who accepted the loan.
accepted_both <- counts["1", "1", "1"]

# Count all customers with CC = 1 and Online = 1.
total_both <- sum(counts["1", , "1"])

# Calculate the acceptance probability for this customer group.
p_direct <- accepted_both / total_both

# Display the numerator, denominator, and probability.
c(Accepted = accepted_both,
  Total = total_both,
  Probability = p_direct)
```

```
##    Accepted       Total Probability
##  57.0000000 532.0000000   0.1071429
```

Counts all customers with a yes for all 3 variables

Counts the CreditCard and Online customers including both loan categories now

Adds the accepted and declined counts for the group

Divides the accepted number by the total number

57 of 532 accepted giving a 10.71% probability of acceptance

```
## Part C creating Two Pivot Tables

# Count loan Responses for each Online Category
loan_online <- table(Loan = train$Loan, Online = train$Online)

# Display Loan Online Table
loan_online
```

```
##     Online
## Loan    0    1
##    0 1102 1620
##    1   99  179
```

```r
# Count Loan Responses for Credit Card Caegory
loan_cc <- table(Loan = train$Loan, CC = train$CC)

# Display the Loan Credit card Table
loan_cc
```

```
##      CC
## Loan    0    1
##    0 1930  792
##    1  187   91
```

The prior gives us the results for Loan Responses in the Online or not category and the Loan responses for Credit Card or not category

We see here that 90% of Online banking customers declined the Loan, and 91.8% of non-online customers declined the loan as well

We also see that 91.2% of non credit-card users declined the Loan, and 89.7% of the Credit-Card users declined the Loan.

```r
## Part D - Calculating probabilities
# i. Credit Card Holders Among Loan Acceptors
p_cc_given_loan1 <- loan_cc["1", "1"] / sum(loan_cc["1", ])

# ii. Online banking users among all loan acceptors
p_online_given_loan1 <- loan_online["1", "1"] / sum(loan_online["1", ])

# iii. Loan acceptors among all training customers
p_loan1 <- sum(train$Loan == "1") / nrow(train)

# iv. Credit Card holders among loan decliners
p_cc_given_loan0 <- loan_cc["0", "1"] / sum(loan_cc["0", ])

# v. Online banking users among loan decliners
p_online_given_loan0 <- loan_online["0", "1"] / sum(loan_online["0", ])

# vi. Loan decliners among all training customers
p_loan0 <- sum(train$Loan == "0") / nrow(train)
```

i. The Probability of Crddit Card given a Loan Acceptance is 32.7%

ii. The Probability of Online Given a Loan Acceptance is 64.4%

iii. The Probability of Loan Acceptors among all training customers is 9.3%

iv. The Probability of Credit Card holder among loan decliners is 29.1%

v. The Probability of Online Given a Loan Decliner is 59.5%

vi. ## The Probability of Loan Decliner among all training customers is 90.7%

```
# Part E - Naive Bayes Probability
## Naive Bayes assumes that Credit Card and Online are independent within Loan category, so we can mult
# Calculate the score for accepting the loan
score1 <- p_cc_given_loan1 * p_online_given_loan1 * p_loan1

# Calculate the score for declining the loan
score0 <- p_cc_given_loan0 * p_online_given_loan0 * p_loan0

# Convert the acceptance into a probability
p_nb_manual <- score1 / (score1 + score0)

# Display the Acceptance Probability
p_nb_manual
```

```
## [1] 0.1105637
```

This formula gives us the total proportion of acceptors, and the total proportion of non acceptors

It then divided the total acceptors by the total acceptors and non acceptors to give us a final probability of (11.1%)

```
# Part F - Comparing the Value with the Pivot Table (Which is more accurate)

# Display the Direct and Naive Bayes Probabilities Together
c(Direct_pivot = p_direct, Naive_Bayes = p_nb_manual)
```

```
## Direct_pivot  Naive_Bayes
##    0.1071429    0.1105637
```

```
# Calculate the Naive Bayes Probability minus direct probability
difference <- p_nb_manual - p_direct

# Display the difference
difference
```

```
## [1] 0.003420806
```

Direct Calculated Probability is 10.7%

Naive Bayes Calculated Probability is 11.1%

The direct estimate matches the training observations well with these conclusions

```
# Part G -

# Train the model to predict Loan using Credit Card and online
# Model that predicts Loan from Credit Card and Online
nb_model <- naiveBayes(Loan ~ CC + Online, data = train, laplace = 0)

# Display the models class counts and conditional probabilities
nb_model
```

```
##
## Naive Bayes Classifier for Discrete Predictors
##
## Call:
## naiveBayes.default(x = X, y = Y, laplace = laplace)
##
## A-priori probabilities:
## Y
##          0          1
## 0.90733333 0.09266667
##
## Conditional probabilities:
##    CC
## Y           0         1
##   0 0.7090375 0.2909625
##   1 0.6726619 0.3273381
##
##    Online
## Y           0         1
##   0 0.4048494 0.5951506
##   1 0.3561151 0.6438849
```

```
# Return the probabilities for each training customer
predicted_train_probabilities <- predict(nb_model, train, type = "raw")

# Display the first six customers predicted probabilities
head(predicted_train_probabilities)
```

```
##              0          1
## [1,] 0.9051206 0.09487943
## [2,] 0.8894363 0.11056366
## [3,] 0.9051206 0.09487943
## [4,] 0.9214655 0.07853447
```

```
## [5,] 0.9051206 0.09487943
## [6,] 0.9214655 0.07853447
```

```r
# Find customers with a bank credit card who use online banking as well
matching_rows <- which(train$CC == "1" & train$Online == "1")

# Display the probabilities for the first matching customer
predicted_train_probabilities[matching_rows[1], ]
```

```
##         0         1
## 0.8894363 0.1105637
```

```r
# Store that customers probability of accepting the loan
p_nb_model <- predicted_train_probabilities[matching_rows[1], "1"]

# Compare the model probability with the manual calculation from Part E
c(Manual = p_nb_manual, Model = p_nb_model)
```

```
##    Manual   Model.1
## 0.1105637 0.1105637
```

```r
# Obtain probabilities for the validation customers
predicted_validation_probabilities <- predict(nb_model, validation, type = "raw")

# Display the first six rows
head(predicted_validation_probabilities)
```

```
##              0         1
## [1,] 0.9214655 0.07853447
## [2,] 0.9214655 0.07853447
## [3,] 0.9051206 0.09487943
## [4,] 0.9082092 0.09179078
## [5,] 0.9051206 0.09487943
## [6,] 0.9214655 0.07853447
```

The model estimates an 11.1% probability of loan acceptance for customers with a bank credit card and use online banking.

A-priori predicts a 9.3% loan acceptance rate from the training set

Among loan acceptors, 32.7% of them had a bank credit card, while 29.1% of non acceptors did. Customers with credit cards are more likely to accept.

Within loan acceptors, 64.4% of them used online banking, and 59.5% declined the loan. Online banking was also used slightly more often with acceptors

The manual and model calculation both returned an acceptance probability of 11.1%.

Part G -

A-priori [P(Loan = 0) = 90.7% / P(Loan = 1) = 9.3%]

CC Table P(CC = 1 | Loan = 0) = 29.1% / P(CC = 1 | Loan = 1) = 32.7%

Online Table P(Online = 1 | Loan = 0) = 59.5% / P(Online = 1 | Loan = 1) = 64.4%

The model uses the Prior Probabilities of Loan = 0 (90.8%), and Loan = 1 (9.3%) along with the probabilities of having a credit card and using online banking within each Loan group. For a customer with a Credit Card and Online banking, the model predicts an 88.9% probability of not accepting, and an 11.1% acceptance probability. This matches the manual naive bayes comparison from Part E.

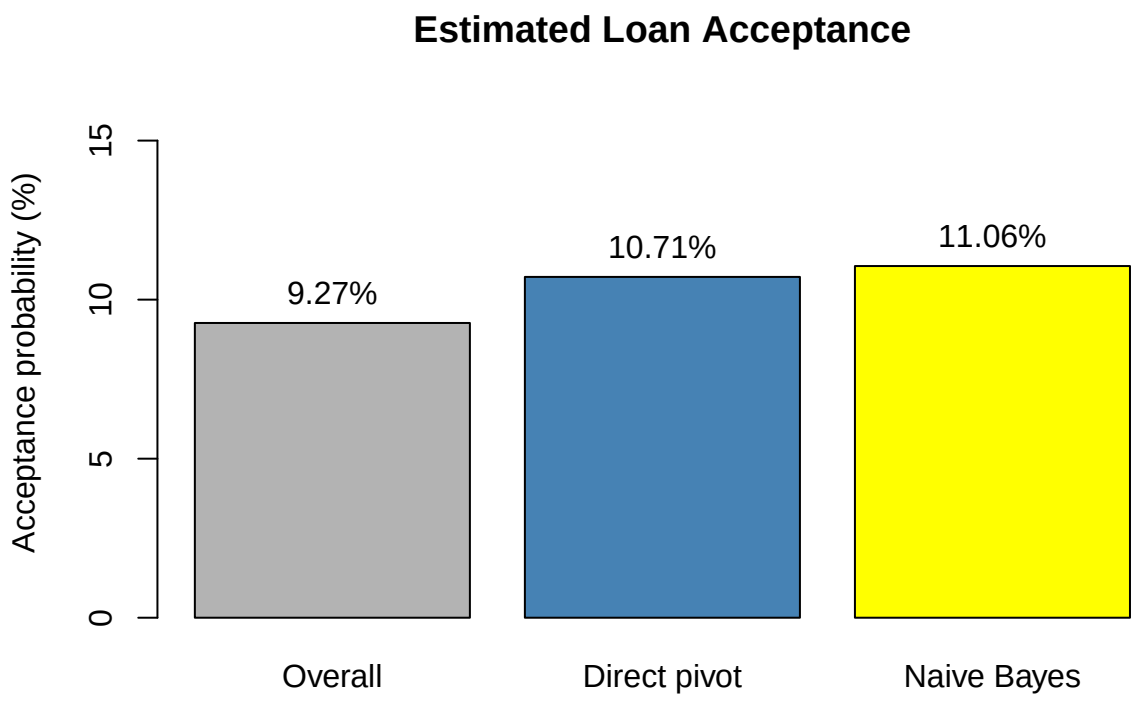
```
# Combine the three acceptance probabilities.
probabilities <- c(p_loan1, p_direct, p_nb_manual)

# Convert probabilities into percentages.
```

```r
percentages <- probabilities * 100

# Draw the bars and save their positions for adding labels.
bar_positions <- barplot(percentages,
                         names.arg = c("Overall", "Direct pivot", "Naive Bayes"),
                         col = c("gray70", "steelblue", "yellow"),
                         ylim = c(0, max(percentages) + 5),
                         main = "Estimated Loan Acceptance",
                         ylab = "Acceptance probability (%)")

# Add the percentage above each bar.
text(x = bar_positions,
     y = percentages,
     labels = paste0(round(percentages, 2), "%"),
     pos = 3)
```

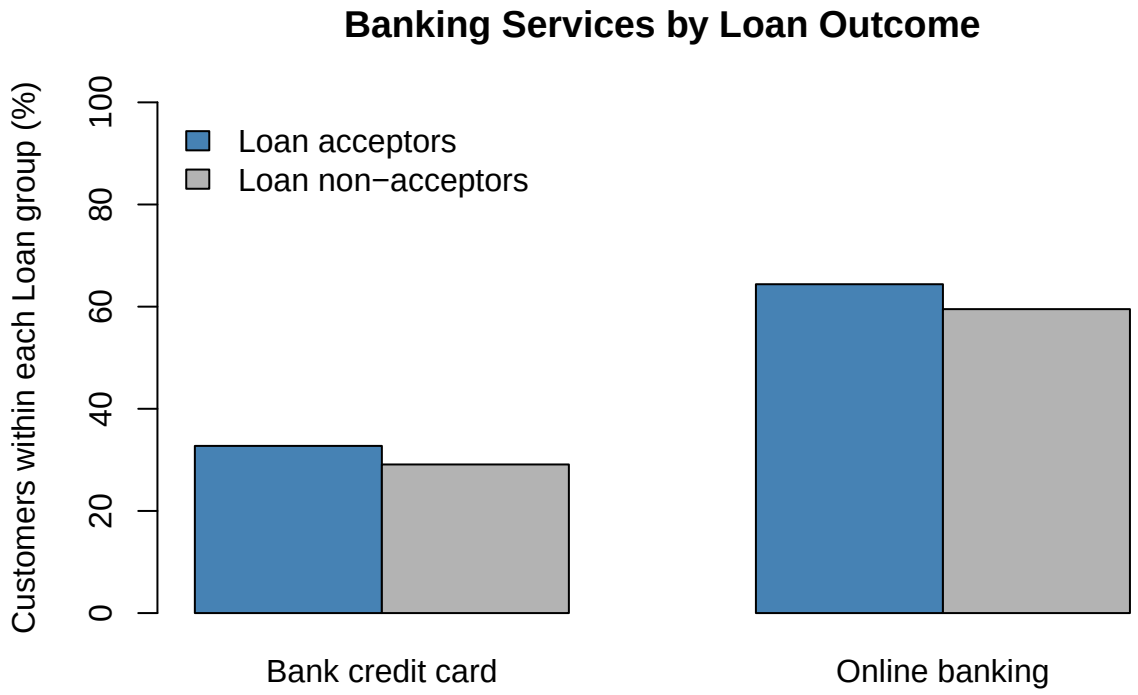


This chart compares the training accpetance loan rate (9.27%), and the direct pivor and Naive Bayes estimated loan acceptance rate for customers with a bank credit card and who use online banking as well. Naive Bayes was 0.35 percentage points higher than the pivot, but shows close approximation to acceptance

```r
# Put the conditional probabilities into a two-row table.
predictor_percentages <- rbind(
  Acceptors = c(p_cc_given_loan1, p_online_given_loan1),
  Non_acceptors = c(p_cc_given_loan0, p_online_given_loan0)
) * 100
```

```r
# Name the two predictor columns.
colnames(predictor_percentages) <- c("Bank credit card", "Online banking")

# Draw acceptor and non-acceptor bars next to each other.
barplot(predictor_percentages,
        beside = TRUE,
        col = c("steelblue", "gray70"),
        ylim = c(0, 100),
        main = "Banking Services by Loan Outcome",
        ylab = "Customers within each Loan group (%)",
        legend.text = c("Loan acceptors", "Loan non-acceptors"),
        args.legend = list(x = "topleft", bty = "n"))
```



Bank Credit Card and Online Banking were more common among loan acceptors than non-acceptors given these graphs, as we can see the slightly larger blue plots than grey.